SONCUYA, Katrina Faye A.

BSChE-2B

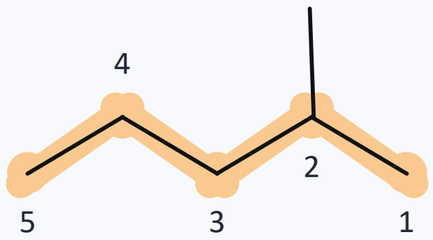
Name It, Draw It

Two-way IUPAC Nomenclature of Alkanes, Cycloalkanes and Haloalkanes

CARD A01 – STRUCTURE – NAME

Foundation · Molecular formula (counted): C6H14

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 5 C → pentane.

Numbering: right → left gives 2; left → right gives 4. 2 < 4, so number from the right.

Substituents: methyl on C2 → 2-methyl.

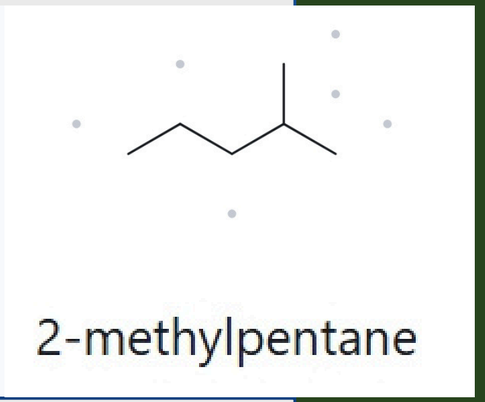
3 · MY NAME (WRITTEN FIRST)

2-methylpentane

4 · THE ALKANE CHECK

Returned name: 2-methylpropane

✓ Match

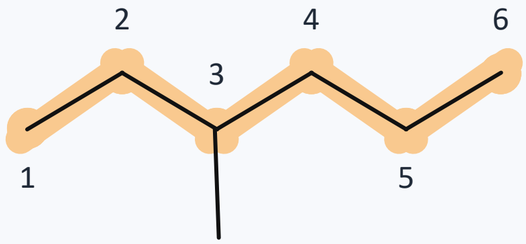


✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A02 – STRUCTURE – NAME

Foundation · Molecular formula (counted): C7H16

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 6 C, running through the ethyl arm (not the horizontal 5 C) → hexane.

Numbering: branch at C3 from one end, C4 from the other. 3 < 4, so number from the end that gives 3.

Substituents: methyl on C3 → 3-methyl.

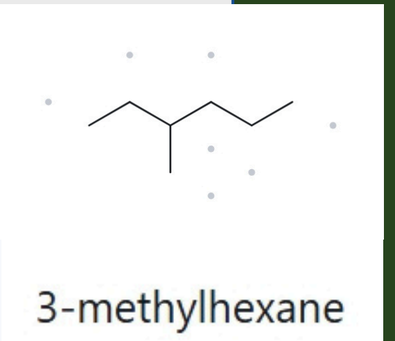
3 · MY NAME (WRITTEN FIRST)

3-methylhexane

4 · THE ALKANE CHECK

Returned name: 3-methylhexane

✓ Match

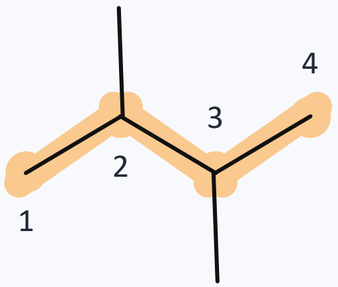


✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A03 – STRUCTURE – NAME

Foundation · Molecular formula (counted): C6H14

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 4 C → butane.

Numbering: both directions give {2,3}: a tie, so either direction works.

Substituents: methyl on C2 and C3 → 2,3-dimethyl.

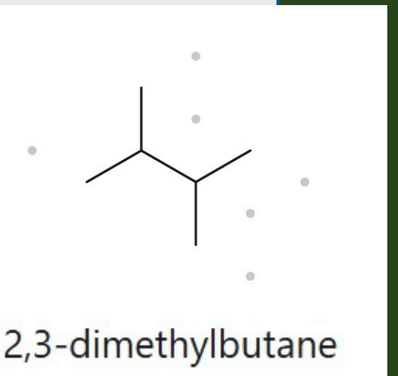
3 · MY NAME (WRITTEN FIRST)

2,3-dimethylbutane

4 · THE ALKANE CHECK

Returned name: 2,3-dimenthylbutane

✓ Match

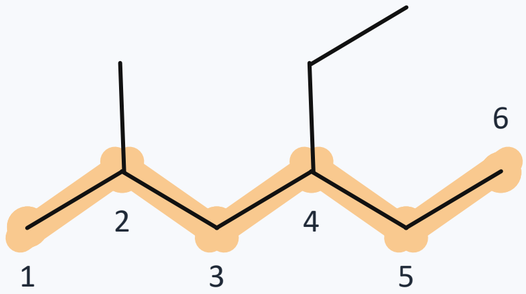


✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A04 – STRUCTURE – NAME

Developing · Molecular formula (counted): C9H20

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 6 C → hexane.

Numbering: left → right gives {2,4}; right → left gives {3,5}. 2 < 3, so number from the left.

Substituents: ethyl on C4, methyl on C2. Ethyl is cited first (e before m).

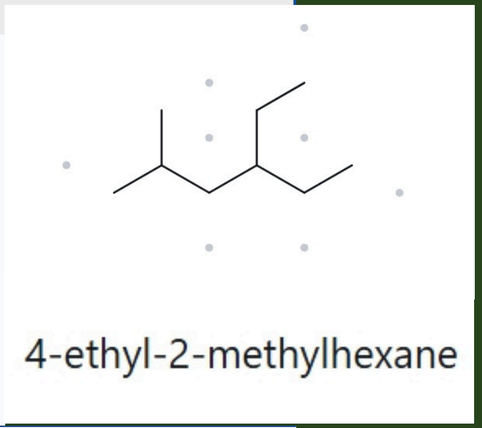
3 · MY NAME (WRITTEN FIRST)

4-ethyl-2-methylhexane

4 · THE ALKANE CHECK

Returned name: 4-ethyl-2-methylhexane

✓ Match

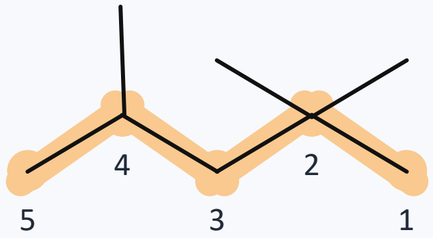


✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A05 – STRUCTURE – NAME

Developing · Molecular formula (counted): C8H18

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 5 C → pentane.

Numbering: right → left gives 2,2,4; left → right gives 2,4,4. First point of difference: 2 < 4, so number from the right.

Substituents: methyl on C2, C2 and C4 → 2,2,4-trimethyl.

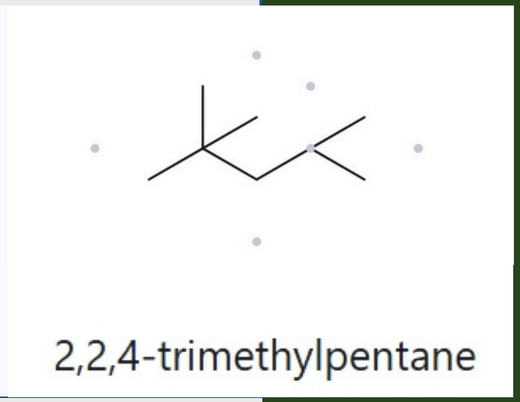
3 · MY NAME (WRITTEN FIRST)

2,2,4-trimethylpentane

4 · THE ALKANE CHECK

Returned name: 2,2,4-trimethylpentane

✓ Match

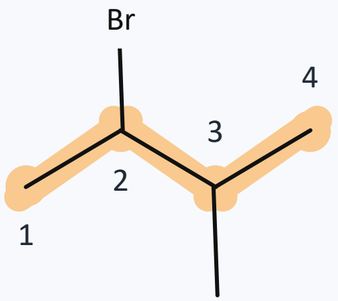


✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A06 – STRUCTURE – NAME

Developing · Molecular formula (counted): C5H11Br

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 4 C → butane.

Numbering: both directions give {2,3}: a tie. Bromo precedes methyl alphabetically, so bromo gets C2.

Substituents: bromo on C2, methyl on C3 → 2-bromo-3-methyl.

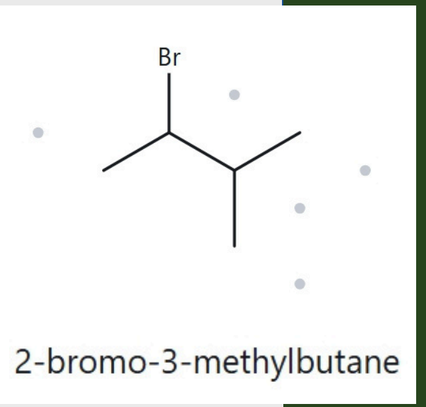
3 · MY NAME (WRITTEN FIRST)

2-bromo-3-methylbutane

4 · THE ALKANE CHECK

Returned name: 2-bromo-3-methylbutane

✓ Match

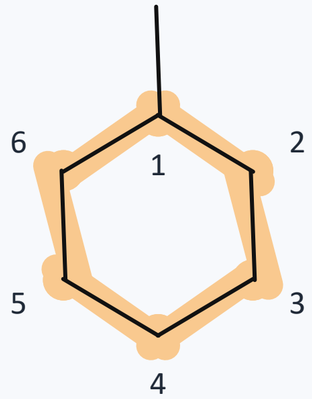


✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A07 – STRUCTURE – NAME

Developing · Molecular formula (counted): C7H14

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: ring of 6 C is the parent → cyclohexane.

Numbering: one substituent only, so no locant is needed.

Substituents: methyl → methyl (no locant).

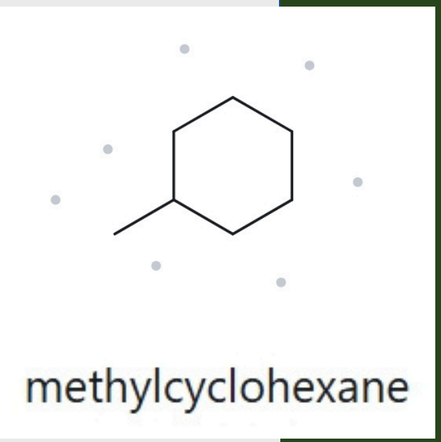
3 · MY NAME (WRITTEN FIRST)

methylcyclohexane

4 · THE ALKANE CHECK

Returned name: methylcyclohexane

✓ Match

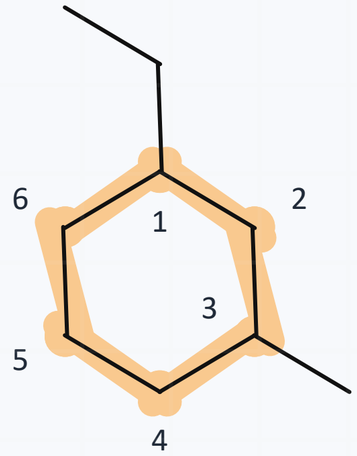


✓ longest chain confirmed **✓ lowest locants** **✓ alphabetical order** **✓ punctuation** **✓ formula counted**

CARD A08 – STRUCTURE – NAME

Challenge · Molecular formula (counted): C9H18

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: ring of 6 C is the parent → cyclohexane.

Numbering: locants are {1,3} either way round the ring: a tie. Ethyl precedes methyl alphabetically, so ethyl gets C1.

Substituents: ethyl on C1, methyl on C3 → 1-ethyl-3-methyl.

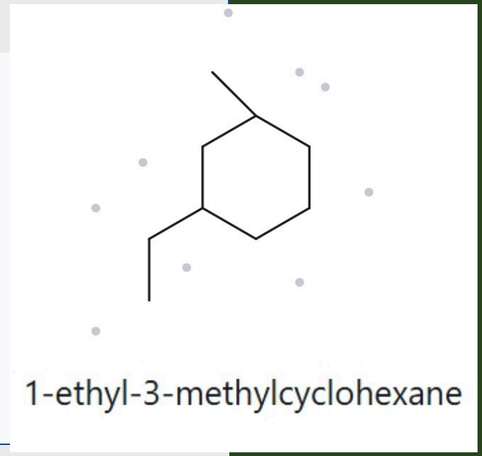
3 · MY NAME (WRITTEN FIRST)

1-ethyl-3-methylcyclohexane

4 · THE ALKANE CHECK

Returned name: 1-ethyl-3-methylcyclohexane

✓ Match

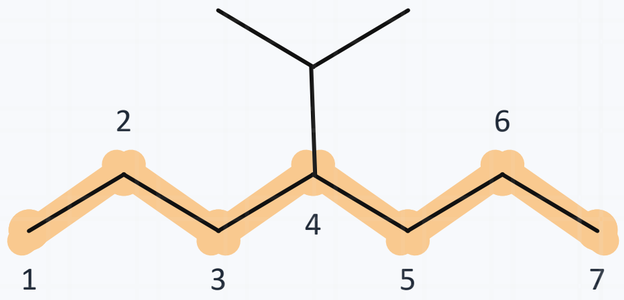


✓ longest chain confirmed **✓ lowest locants** **✓ alphabetical order** **✓ punctuation** **✓ formula counted**

CARD A09 – STRUCTURE – NAME

Challenge · Molecular formula (counted): C10H22

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 7 C → heptane. The path through the branch is only 6 C.

Numbering: the branch is at C4 from either end.

Substituents: isopropyl on C4 (also written (1-methylethyl) or propan-2-yl).

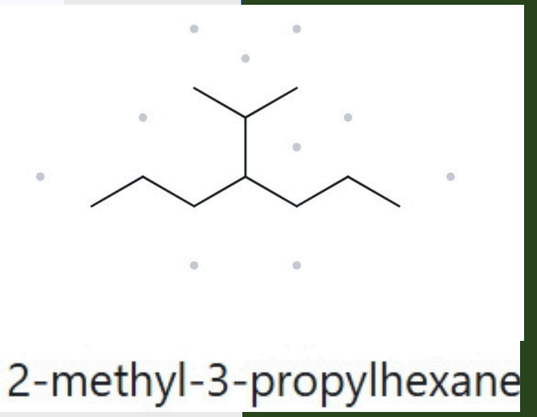
3 · MY NAME (WRITTEN FIRST)

4-isopropylheptane

4 · THE ALKANE CHECK

Returned name: 2-methyl-3-propylhexane

✗ Mismatch (state the correct name and the rule)

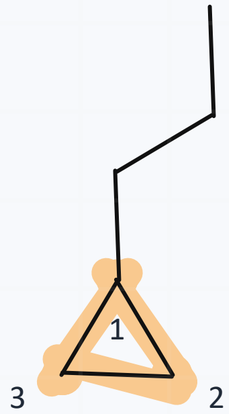


✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD A10 – STRUCTURE – NAME

Challenge · Molecular formula (counted): C6H12

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: ring of 3 C is the parent → cyclopropane (this activity prefers the ring).

Numbering: one substituent only, so no locant is needed.

Substituents: propyl → propyl (no locant).

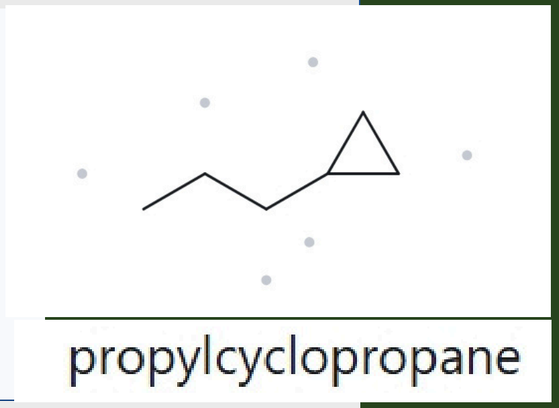
3 · MY NAME (WRITTEN FIRST)

propylcyclopropane

4 · THE ALKANE CHECK

Returned name: propylcyclopropane

✓ Match



✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

CARD B01-NAME-STRUCTURE

Foundation · Given name: 2-methylpropane

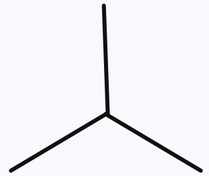
1 · NAME DECODED

Parent: propane → a 3-carbon chain.

Substituents: methyl on C2.

Expected formula: 3 + 1 = 4 C; open chain → C4H10

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C4H10

CnH2n+2 = C4H10 ✓

4 · BACK-CHECK

Returned by The Alkane: 2-methylpropane

✓ Same as the given name

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B02-NAME-STRUCTURE

Foundation · Given name: 3-ethylpentane

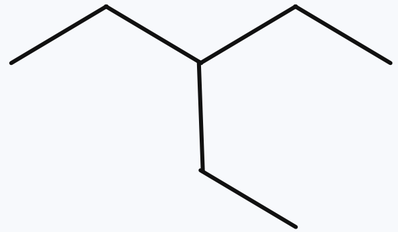
1 · NAME DECODED

Parent: pentane → a 5-carbon chain.

Substituents: ethyl on C3.

Expected formula: 5 + 2 = 7 C; open chain → C7H16

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C7H16

CnH2n+2 = C7H16 ✓

4 · BACK-CHECK

Returned by The Alkane: 3-ethylpentane

✓ Same as the given name

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B03 – NAME → STRUCTURE

Foundation · Given name: 2,2-dimethylbutane

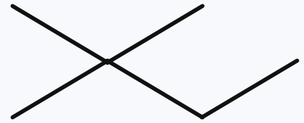
1 · NAME DECODED

Parent: butane → a 4-carbon chain.

Substituents: two methyls, both on C2.

Expected formula: 4 + 2 = 6 C; open chain → C6H14

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C6H14

CnH2n+2 = C6H14 ✓

4 · BACK-CHECK

Returned by The Alkane: 2,2-dimethylbutane

✓ Same as the given name

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

CARD B04-NAME-STRUCTURE

Developing · Given name: 3-ethyl-2-methylpentane

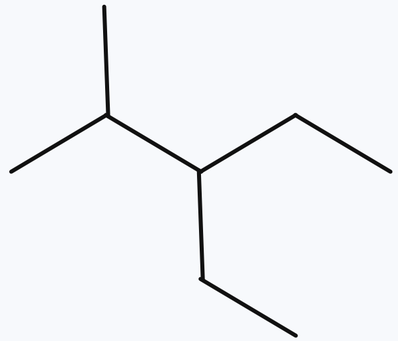
1 · NAME DECODED

Parent: pentane → a 5-carbon chain.

Substituents: methyl on C2; ethyl on C3.

Expected formula: 5 + 1 + 2 = 8 C; open chain → C8H18

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C8H18

CnH2n+2 = C8H18 ✓

4 · BACK-CHECK

Returned by The Alkane: 3-ethyl-2-methylpentane

✓ Same as the given name

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B05-NAME-STRUCTURE

Developing · Given name: 1-chloro-2-methylpropane

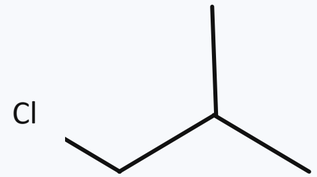
1 · NAME DECODED

Parent: propane → a 3-carbon chain.

Substituents: chloro on C1; methyl on C2.

Expected formula: 3 + 1 = 4 C; C4H10, Cl replaces one H → C4H9Cl

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C4H9Cl

C4H10 minus one H for Cl = C4H9Cl ✓

4 · BACK-CHECK

Returned by The Alkane: 1-chloro-2-methylpropane

✓ Same as the given name

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B06-NAME-STRUCTURE

Developing · Given name: 1,2-dimethylcyclopentane

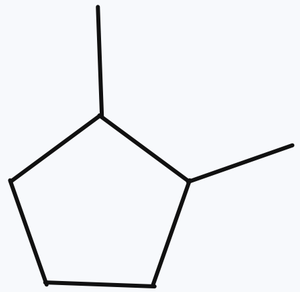
1 · NAME DECODED

Parent: cyclopentane → a 5-carbon ring.

Substituents: methyls on C1 and C2.

Expected formula: 5 + 2 = 7 C; one ring → C7H14

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C7H14

One ring: CnH2n = C7H14 ✓

4 · BACK-CHECK

Returned by The Alkane: 1,2-dimethylcyclopentane

✓ Same as the given name

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B07-NAME-STRUCTURE

Developing · Given name: 4-ethyl-3,3-dimethylheptane

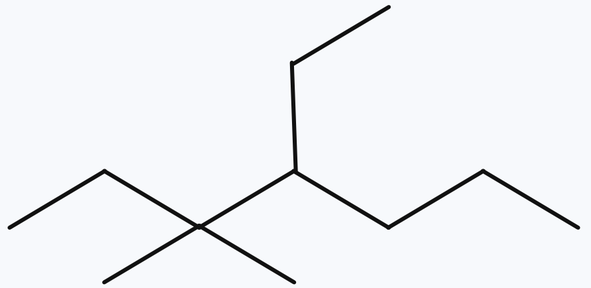
1 · NAME DECODED

Parent: heptane → a 7-carbon chain.

Substituents: two methyls on C3; ethyl on C4.

Expected formula: 7 + 2 + 2 = 11 C; open chain → C11H24

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C11H24

CnH2n+2 = C11H24 ✓

4 · BACK-CHECK

Returned by The Alkane: 4-ethyl-3,3-dimethylheptane

✓ Same as the given name

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B08 – NAME → STRUCTURE

Challenge · Given name: tert-butylcyclohexane

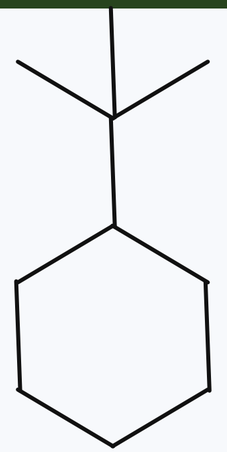
1 · NAME DECODED

Parent: cyclohexane → a 6-carbon ring.

Substituents: tert-butyl, C(CH3)3, on C1.

Expected formula: 6 + 4 = 10 C; one ring → C10H20

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C10H20

One ring: CnH2n = C10H20 ✓

4 · BACK-CHECK

Returned by The Alkane: (1,1-dimethylethyl)cyclohexane (expected).
Same compound: tert-butyl and (1,1-dimethylethyl) are equivalent forms.

✓ Same compound as given

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B09 – NAME → STRUCTURE

Challenge · Given name: 3-ethyl-1,1-dimethylcyclohexane

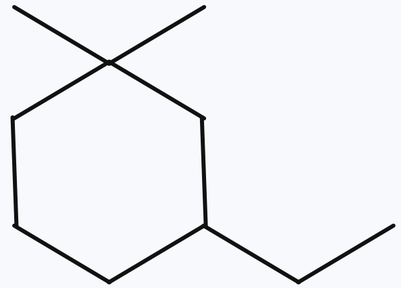
1 · NAME DECODED

Parent: cyclohexane → a 6-carbon ring.

Substituents: two methyls on C1; ethyl on C3.

Expected formula: 6 + 2 + 2 = 10 C; one ring → C10H20

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C10H20

One ring: CnH2n = C10H20 ✓

4 · BACK-CHECK

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane

✓ Same as the given name

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

CARD B10 – NAME → STRUCTURE

Challenge · Given name: 2-ethylbutane (deliberately incorrect)

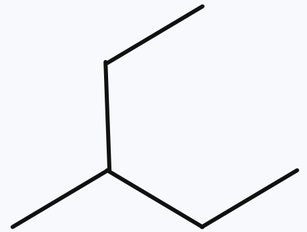
1 · NAME DECODED

Parent: butane → a 4-carbon chain, drawn exactly as written.

Substituents: ethyl on C2.

Expected formula: 4 + 2 = 6 C; open chain → C6H14

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C6H14

CnH2n+2 = C6H14 ✓

4 · BACK-CHECK

Returned by The Alkane: 3-methylpentane (expected). The ethyl branch extends the chain to 5 C, so the parent is pentane with methyl on C3. Rule broken: the longest chain was not chosen as the parent.

✗ Differs from the given name

✓ parent length matches **✓ substituent positions match** **✓ formula agrees** **✓ back-check recorded**

SONCUYA, Katrina Faye A.

BSChE-2B

Thank You